
Verification-Gated Autonomous Optimization of a Production Raft Library: A Harness-First Pipeline for Algorithmic Discovery

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We study whether an autonomous, LLM-driven optimization loop can discover
2 algorithmic improvements to a production consensus library while preserving every
3 protocol-level correctness invariant. Taking `etcd-io/raft`—the most widely
4 deployed Raft implementation—as our target, we build a layered verification
5 harness combining the in-tree TLA^+ specification, deterministic simulation testing
6 under seeded fault injection, linearizability checking of recorded client histories,
7 and a differential oracle against the unmodified upstream library. We then drive an
8 evolutionary search in which a large language model proposes bounded algorithmic
9 edits and configuration variants; candidates that fail any harness layer are rejected
10 before any performance comparison is permitted. We report (i) the design and
11 injection points of the harness over Raft’s existing `Node/Storage` interfaces, (ii) the
12 set of behavioural anomalies surfaced during harness bring-up, (iii) the candidate
13 variants accepted by the harness and their throughput and tail-latency characteristics
14 relative to the pinned baseline, and (iv) an ablation showing which harness layer
15 caught which class of incorrect candidate. Our central claim is methodological:
16 in a domain with mechanically checkable safety properties, the harness—not the
17 model—is the load-bearing artefact, and a sufficiently sharp harness lets a general-
18 purpose coding agent search the algorithmic neighbourhood of a mature systems
19 codebase safely.

20	1	Introduction
21	2	Background
22	3	Verification Harness
23	3.1	Design Principles
24	3.2	Layer 1: TLA⁺ Specification
25	3.3	Layer 2: Deterministic Simulation Testing
26	3.4	Layer 3: Linearizability
27	3.5	Layer 4: Differential Oracle
28	3.6	Injection Points and Costs
29	4	Autonomous Optimization Loop
30	4.1	Search Space
31	4.2	Candidate Generator
32	4.3	Rejection Oracle
33	4.4	Candidate Database and Selection
34	5	Experimental Setup
35	6	Results
36	6.1	Harness Bring-Up Findings
37	6.2	Accepted Variants
38	6.3	Per-Layer Ablation
39	6.4	Cost Analysis
40	7	Limitations and Threats to Validity
41	8	Related Work
42	9	Conclusion
43		References
44	A	Technical Appendices and Supplementary Material

45 Conference For AI Scientists 2026 - AI Involvement Checklist

46 This checklist is designed to allow you to explain the role of AI in your research. This is important for
47 understanding broadly how researchers use AI and how this impacts the quality and characteristics of
48 the research.

49 The checklist has two parts: **Research Stage Assessment** (items 1-4), where you rate the level of AI
50 involvement and iteration effort at each stage of the research process, and **AI System Documentation**
51 (items 5-6), where you provide free-text descriptions of the AI systems used and their observed
52 limitations.

53 Research Stage Assessment

54 For items 1-4, give a score from the scale below that defines the role of AI in each part of the scientific
55 process. The scores are as follows:

- 56 • **[A] Human-generated:** Humans generated 95% or more of the research, with AI being of
57 minimal involvement.
- 58 • **[B] Mostly human, assisted by AI:** The research was a collaboration between humans and
59 AI models, but humans produced the majority (>50%) of the research.
- 60 • **[C] Mostly AI, assisted by human:** The research task was a collaboration between humans
61 and AI models, but AI produced the majority (>50%) of the research.
- 62 • **[D] AI-generated:** AI performed over 95% of the research. This may involve minimal
63 human involvement, such as prompting or high-level guidance during the research process,
64 but the majority of the ideas and work came from the AI.

65 For each research stage where AI was involved, i.e. where you selected **[B]** , **[C]** , or **[D]** , please
66 also indicate the approximate level of iteration effort required using the provided iteration macros.
67 Count substantive attempts, prompts, runs, agent trajectories, or course corrections; do not count
68 minor wording edits.

- 69 • **[Low]** : approximately 1-10 substantive attempts.
- 70 • **[Medium]** : approximately 10-100 substantive attempts, with some failed attempts or course
71 corrections.
- 72 • **[High]** : approximately 100+ substantive attempts, with substantial exploration, failures, or
73 trial-and-error.
- 74 • **[NA]** : not applicable because AI involvement was **[A]** .
- 75 • **[Unclear]** : the authors cannot reasonably estimate.

76 These categories leave room for interpretation, so we ask that the authors also include a brief
77 explanation elaborating on how AI was involved in the tasks for each category. Please keep your
78 explanation to less than 150 words.

79 1. **Hypothesis development:** Hypothesis development includes the process by which you
80 came to explore this research topic and research question. This can involve the background
81 research performed by either researchers or by AI. This can also involve whether the idea
82 was proposed by researchers or by AI.

83 Answer: **[TODO]**

84 Iteration Effort: **[TODO]**

85 Explanation: **[TODO]**

86 2. **Experimental design and implementation:** This category includes design of experiments
87 that are used to test the hypotheses, coding and implementation of computational methods,
88 and the execution of these experiments.

89 Answer: **[TODO]**

90 Iteration Effort: **[TODO]**

91 Explanation: **[TODO]**

- 92 3. **Analysis of data and interpretation of results:** This category encompasses any process to
93 organize and process data for the experiments in the paper. It also includes interpretations of
94 the results of the study.
95 Answer: [TODO]
96 Iteration Effort: [TODO]
97 Explanation: [TODO]
- 98 4. **Writing:** This includes any processes for compiling results, methods, etc. into the final
99 paper form. This can involve not only writing of the main text but also figure-making,
100 improving layout of the manuscript, and formulation of narrative.
101 Answer: [TODO]
102 Iteration Effort: [TODO]
103 Explanation: [TODO]

104 **AI System Documentation**

105 For items 5-6, provide a free-text description.

- 106 5. **AI systems used:** Describe all AI systems used in this research without naming specific
107 products or models. Use system-level descriptors only (e.g., “a frontier large language
108 model”, “a protein structure prediction system”, “a custom multi-agent framework built on
109 open-source models”). Include relevant details such as system type, scale, and capabilities.
110 **Specific model and product names should only be included in the camera-ready version.**
111 Description: [TODO]
- 112 6. **Observed AI Limitations:** What specific limitations and failure modes have you observed
113 when using AI as a partner or lead author?
114 Description: [TODO]

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: **[TODO]**

Justification: **[TODO]**

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: **[TODO]**

Justification: **[TODO]**

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: **[TODO]**

Justification: **[TODO]**

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: **[TODO]**

Justification: **[TODO]**

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: **[TODO]**

Justification: **[TODO]**

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

9. Research Ethics

161 Question: Does the research conducted in the paper conform with the NeurIPS Code of
162 Ethics, except where CAISc's AI authorship policies apply?
163 Answer: **[TODO]**
164 Justification: **[TODO]**
165 10. **Broader impacts**
166 Question: Does the paper discuss both potential positive societal impacts and negative
167 societal impacts of the work performed?
168 Answer: **[TODO]**
169 Justification: **[TODO]**